

Module 4

Theory of Reference Frames (Chap. 3 of Krause's Book)

Most Important Topics and Concepts

□ Principles of:

- Advantages of reference frame transformations
- Stationary and rotating reference frames
- qd and $\alpha\beta$ coordinates
- Modeling of circuits in arbitrary reference frames
- Special cases of arbitrary reference frames
- Transformation between reference frames

it does not rotate

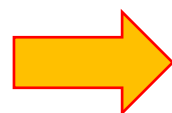
general

Motivation for Transformations

❖ Solving State of the System:

➤ Consider a Linear Time-Varying (LTV) System:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u} \\ \mathbf{y} = \mathbf{C}(t)\mathbf{x} + \mathbf{D}(t)\mathbf{u} \end{cases}$$



• is it easy to solve? **NO!**

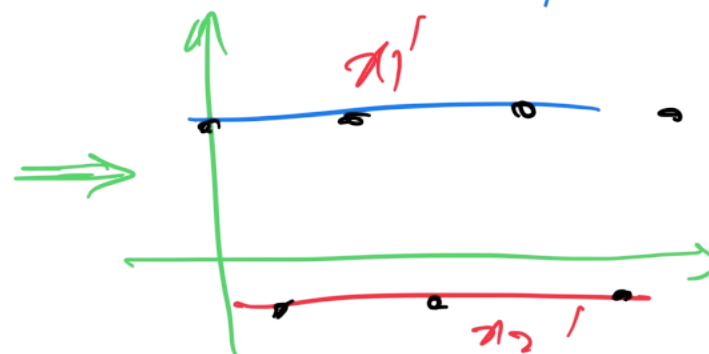
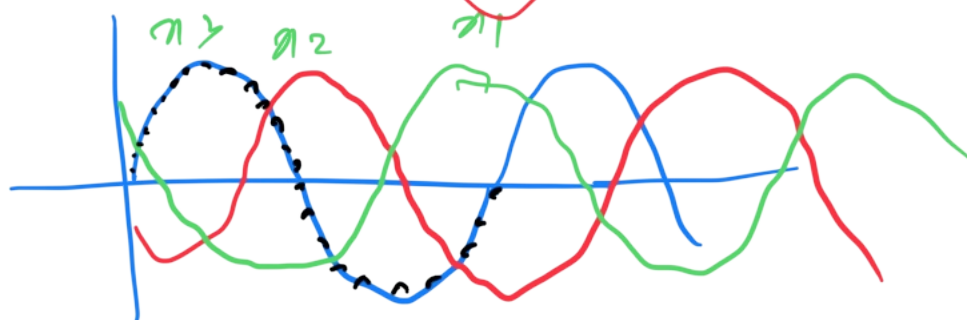
because of time-varying matrices

- Is the choice of state variables in $\mathbf{x}$ unique?

NO!

$$\mathbf{y} = \mathbf{f}(\mathbf{x})$$

solve for $\mathbf{x} = \mathbf{f}^{-1}(\mathbf{y})$



Transformation of Variables

➤ Define New States for the System:

transformation matrix

original state vector

$$\tilde{\mathbf{x}} = \mathbf{K}(t) \mathbf{x}$$

new state vector

Assume $\mathbf{K}(t) \in R^{n \times n}$ and $\mathbf{K}$ is invertible

$$\mathbf{x} = \mathbf{K}^{-1}(t) \tilde{\mathbf{x}}$$

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u} \\ \mathbf{y} = \mathbf{C}(t)\mathbf{x} + \mathbf{D}(t)\mathbf{u} \end{cases}$$

$$\frac{d(\mathbf{K}^{-1}(t) \tilde{\mathbf{x}})}{dt} = \mathbf{A}(t) \mathbf{K}^{-1}(t) \tilde{\mathbf{x}} + \mathbf{B}(t) \mathbf{u}$$

$$\mathbf{y} = \mathbf{C}(t) \mathbf{K}^{-1}(t) \tilde{\mathbf{x}} + \mathbf{D}(t) \mathbf{u}$$

$$(\mathbf{f} \cdot \mathbf{g})' = \mathbf{f}' \cdot \mathbf{g} + \mathbf{f} \cdot \mathbf{g}'$$

$$\frac{d(\mathbf{K}^{-1}(t) \tilde{\mathbf{x}})}{dt} + (\mathbf{K}^{-1}(t) \frac{d\tilde{\mathbf{x}}}{dt}) = \left(\mathbf{A}(t) \mathbf{K}^{-1}(t) \tilde{\mathbf{x}} + \mathbf{B}(t) \mathbf{u} - \frac{d(\mathbf{K}^{-1}(t))}{dt} \tilde{\mathbf{x}} \right)$$

$$\frac{d\tilde{\mathbf{x}}}{dt} = \underbrace{\left[\mathbf{K}(t) \mathbf{A}(t) \mathbf{K}^{-1}(t) - \mathbf{K}(t) \frac{d(\mathbf{K}^{-1}(t))}{dt} \right]}_{\tilde{\mathbf{A}} - \mathbf{W}} \tilde{\mathbf{x}} + \underbrace{\mathbf{K}(t) \mathbf{B}(t)}_{\tilde{\mathbf{B}}} \mathbf{u}$$

Transformation of Variables

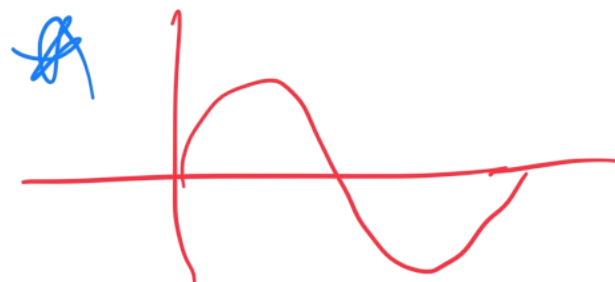
➤ Define New State Equations for System:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u} \\ \mathbf{y} = \mathbf{C}(t)\mathbf{x} + \mathbf{D}(t)\mathbf{u} \end{cases}$$

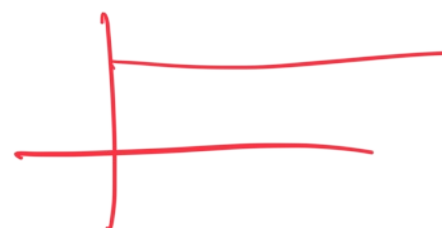
$$\tilde{\mathbf{x}} = \mathbf{K}(t)\mathbf{x}$$

$$\begin{cases} \tilde{\mathbf{A}} = \mathbf{K}(t)\mathbf{A}(t)\mathbf{K}^{-1}(t) \\ \mathbf{W} = \mathbf{K}(t)\frac{d}{dt}(\mathbf{K}^{-1}(t)) \\ \tilde{\mathbf{B}} = \mathbf{K}(t)\mathbf{B}(t) \\ \tilde{\mathbf{C}} = \mathbf{K}^{-1}(t)\mathbf{C}(t) \end{cases}$$

$$\begin{cases} \frac{d\tilde{\mathbf{x}}}{dt} = (\tilde{\mathbf{A}} - \mathbf{W})\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\mathbf{u} \\ \mathbf{y} = \tilde{\mathbf{C}}\tilde{\mathbf{x}} + \mathbf{D}\mathbf{u} \end{cases}$$



$\Rightarrow$



$\tilde{\mathbf{A}} - \mathbf{W}$: constant

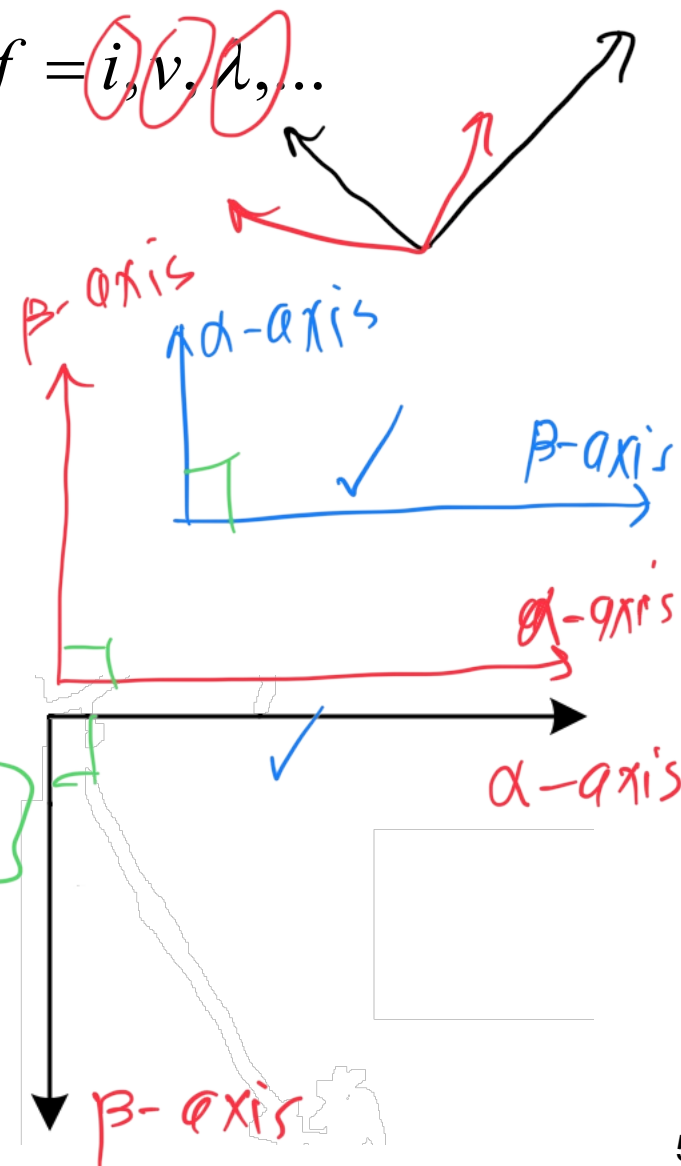
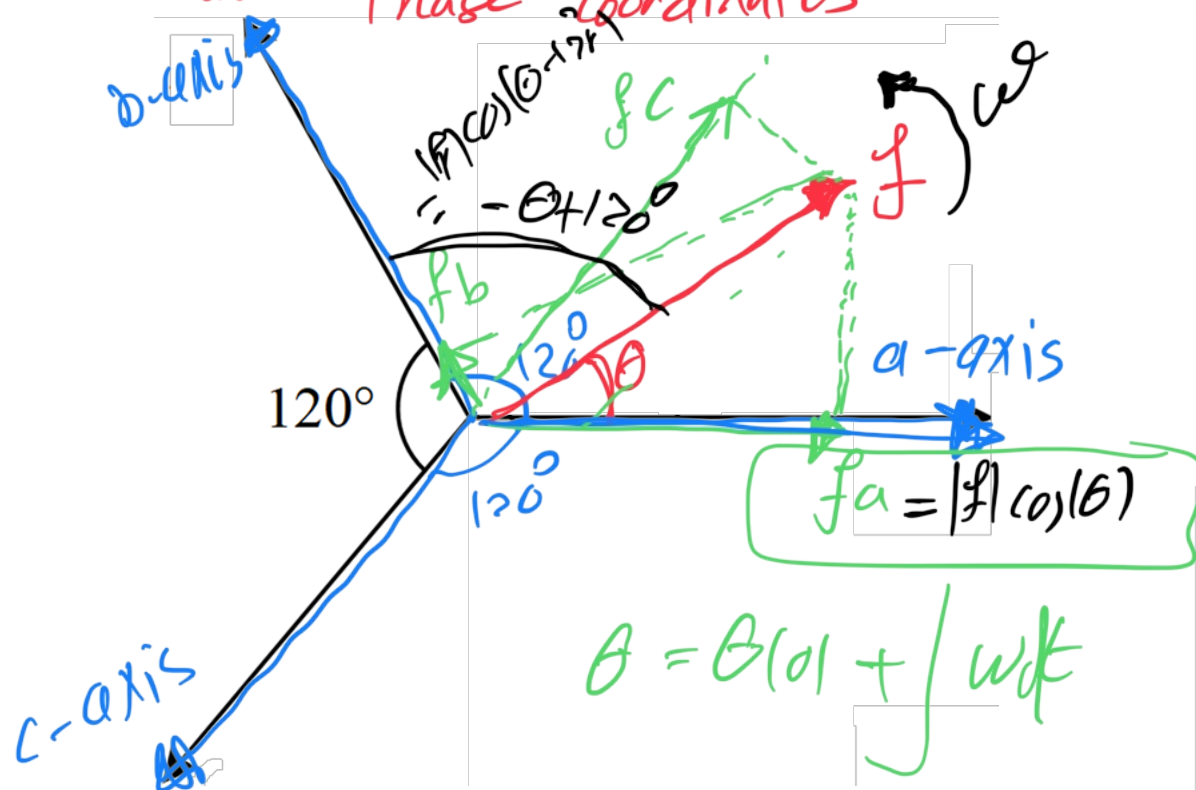
Reference Frames for 3-Phase Systems

❖ Stationary Reference Frames:

- Consider that each phase has variables $f = i, v, \lambda, \dots$

Define a vector $\mathbf{f}_{abc} = [f_a \ f_b \ f_c]^T$

abc Phase coordinates

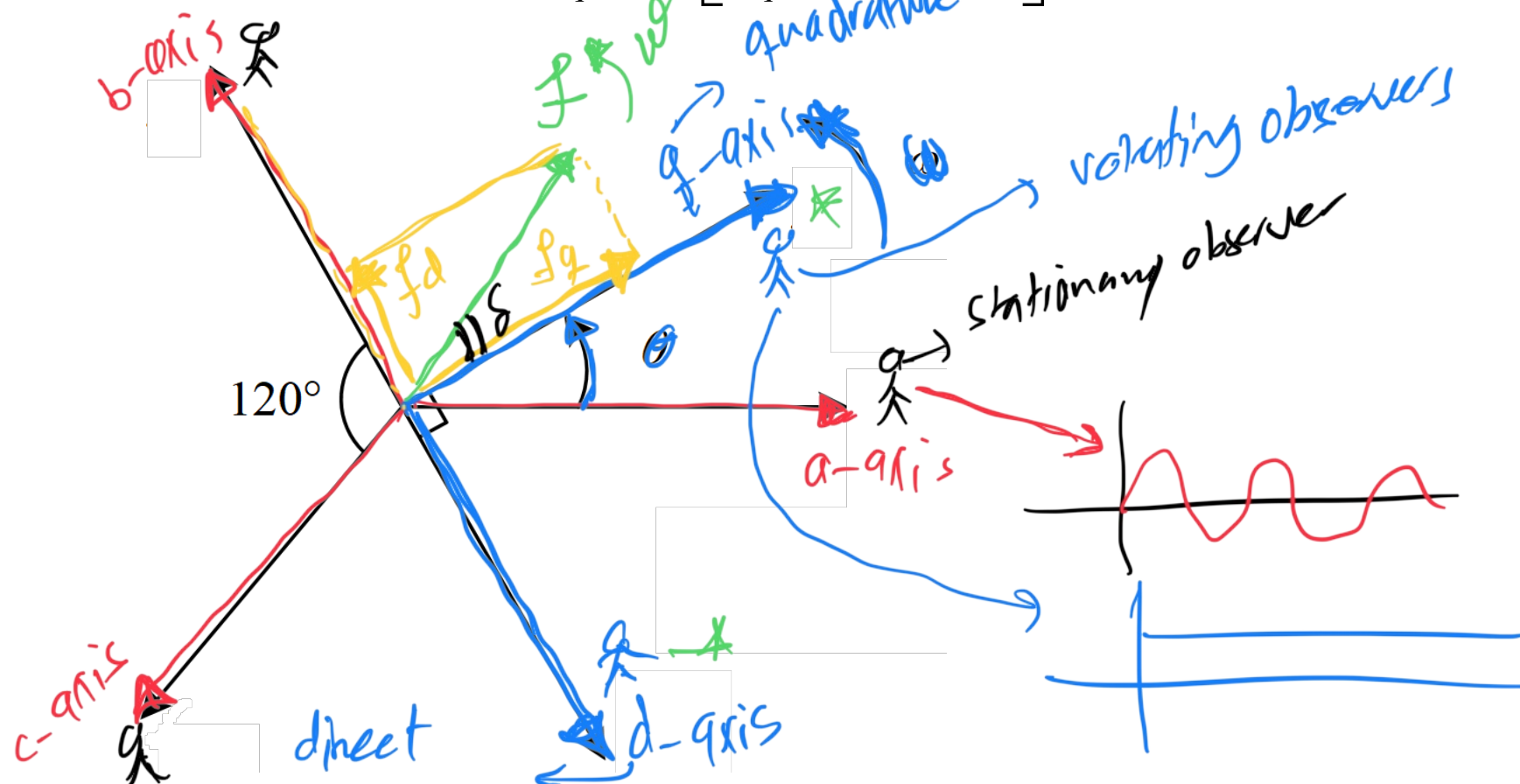


Reference Frames for 3-Phase Systems

❖ Rotating Reference Frames:

- Consider that each phase has variables $f = i, v, \lambda, \dots$

Define a new vector $\mathbf{f}_{qd0} = \begin{bmatrix} f_q & f_d & f_0 \end{bmatrix}^T$



Reference Frames for 3-Phase Systems

❖ Arbitrary Reference Frame (ARF):

➤ Consider that each phase has variables $f = i, v, \lambda, \dots$

$$\begin{cases} \mathbf{f}_{abc} = [f_a \ f_b \ f_c]^T \\ \mathbf{f}_{qd0} = [f_q \ f_d \ f_0]^T \end{cases}$$

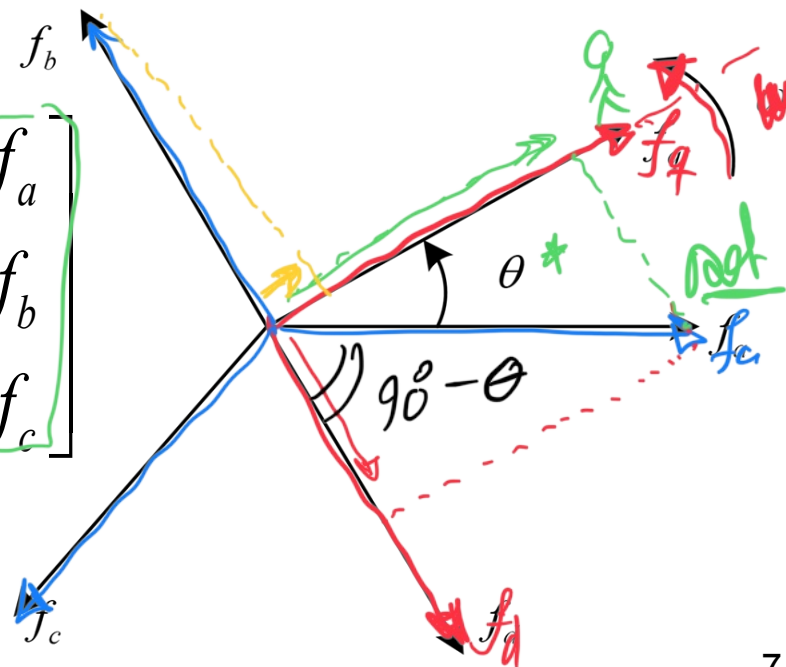
Define a
Transformation Matrix

$$\begin{cases} \mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc} \\ \mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0} \end{cases}$$

$$\begin{bmatrix} f_q \\ f_d \\ f_0 \end{bmatrix} = \left(\frac{2}{3} \right) \begin{bmatrix} \cos(\theta) & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin(\theta) & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix}$$

3x3

Assume peaks of **qd** = peaks of **abc**



Reference Frames for 3-Phase Systems

❖ Arbitrary Reference Frame (ARF):

➤ Consider that each phase has variables $f = i, v, \lambda, \dots$

$$\begin{cases} \mathbf{f}_{abc} = [f_a & f_b & f_c]^T \\ \mathbf{f}_{qd0} = [f_q & f_d & f_0]^T \end{cases}$$

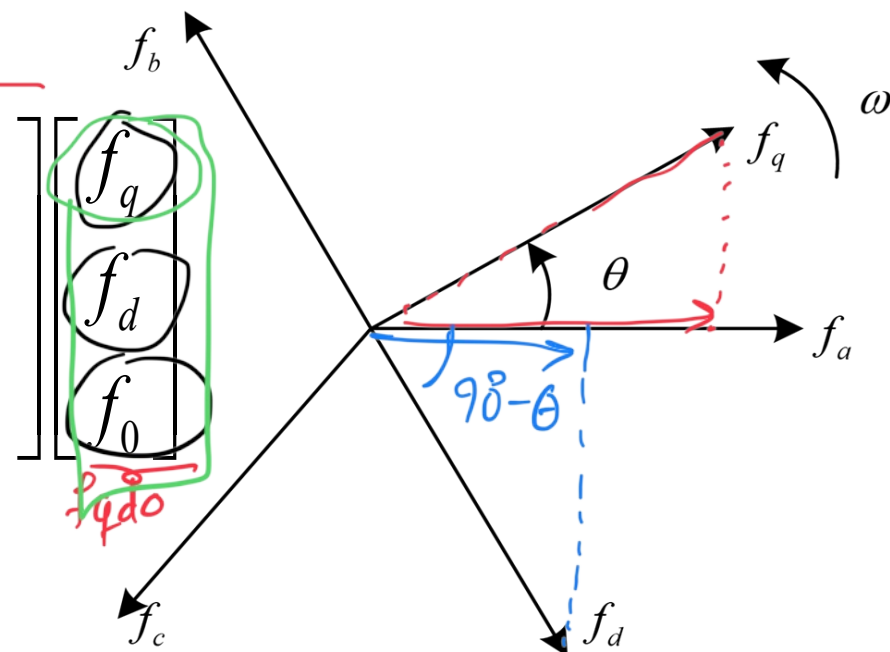
Define a
Transformation Matrix

$$\begin{cases} \mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc} \\ \mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0} \end{cases}$$

$$\begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 1 \\ \cos(\theta - 120^\circ) & \sin(\theta - 120^\circ) & 1 \\ \cos(\theta + 120^\circ) & \sin(\theta + 120^\circ) & 1 \end{bmatrix} \begin{bmatrix} f_q \\ f_d \\ f_0 \end{bmatrix}$$

$\mathbf{f}_{abc}$

Assume peaks of qd = peaks of abc



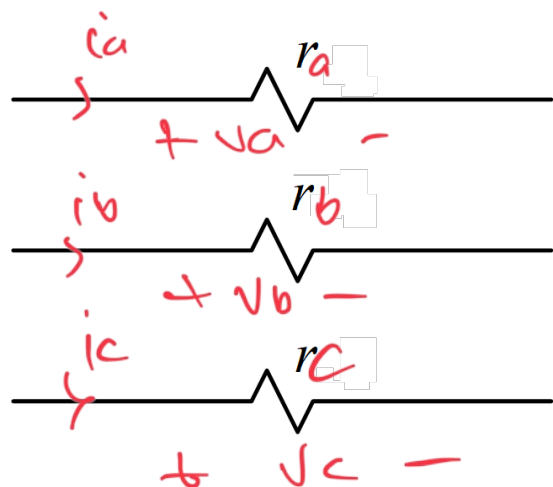
$$\underbrace{\left(\frac{2}{3} \right) \begin{bmatrix} \cos \theta & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin \theta & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}}_{K_s} \underbrace{\begin{bmatrix} \cos \theta & \sin \theta \\ \cos(\theta - 120^\circ) & \sin(\theta - 120^\circ) \\ \cos(\theta + 120^\circ) & \sin(\theta + 120^\circ) \end{bmatrix}}_{K_s^{-1}} =$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}$$

$$\left(\frac{2}{3} \right) (1) \underbrace{\left[\cos^2 \theta + \cos^2(\theta - 120^\circ) + \cos^2(\theta + 120^\circ) \right]}_{\frac{3}{2}} = 1$$

Transforming Circuit Variables to ARF

❖ Resistive Circuits:



$$\mathbf{v}_{abc} = \mathbf{r}_{abc} \mathbf{i}_{abc}$$

$$\mathbf{r}_{abc} = \begin{bmatrix} r_a & 0 & 0 \\ 0 & r_b & 0 \\ 0 & 0 & r_c \end{bmatrix}$$

$$\mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc}$$

$$\mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0}$$

$$[\mathbf{K}_s^{-1}] \mathbf{v}_{qd0} = \mathbf{v}_{abc} \quad ([\mathbf{K}_s] \mathbf{i}_{qd0})$$

$$\mathbf{v}_{qd0} = (\mathbf{K}_s \mathbf{r}_{abc} \mathbf{K}_s^{-1}) \mathbf{i}_{qd0}$$

$$\mathbf{r}_{qd0} = \mathbf{K}_s \begin{bmatrix} r_a & & \\ & r_b & \\ & & r_c \end{bmatrix} \mathbf{K}_s^{-1}$$

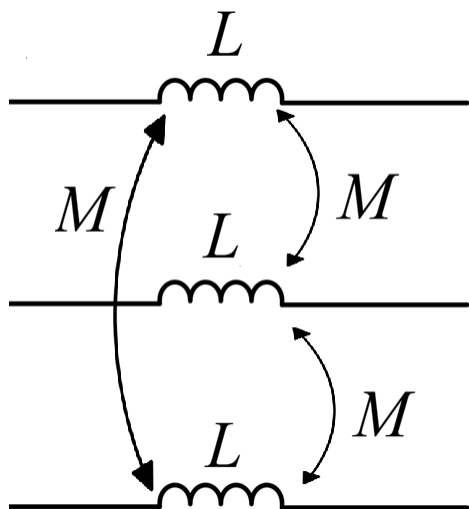
If $r_a = r_b = r_c = r$

$$\mathbf{r}_{qd0} = \mathbf{K}_s r \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \mathbf{K}_s^{-1} =$$

$$\mathbf{r}_{qd0} = r \mathbf{K}_s \mathbf{K}_s^{-1} = r \mathbf{I}_3$$

Transforming Circuit Variables to ARF

❖ Inductive Circuits:



$$\mathbf{v}_{abc} = \frac{d\boldsymbol{\lambda}_{abc}}{dt}$$

$$\boldsymbol{\lambda}_{abc} = \mathbf{L}_{abc} \mathbf{i}_{abc}$$

$$\mathbf{L}_{abc} = \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix}$$

$$\mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc}$$

$$\mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0}$$

$$[\mathbf{K}_s^{-1} \mathbf{v}_{qd0}] = \frac{d}{dt} [\mathbf{K}_s^{-1} \boldsymbol{\lambda}_{qd0}]$$

$$(\mathbf{f} \cdot \mathbf{g})' = \mathbf{f} \cdot \mathbf{g}' + \mathbf{f}' \cdot \mathbf{g}$$

$$\mathbf{K}_s^{-1} \mathbf{v}_{qd0} = \mathbf{K}_s^{-1} \cdot \frac{d}{dt} (\boldsymbol{\lambda}_{qd0}) + \frac{d}{dt} (\mathbf{K}_s^{-1}) \boldsymbol{\lambda}_{qd0}$$

$$\mathbf{v}_{qd0} = \frac{d(\boldsymbol{\lambda}_{qd0})}{dt} + \mathbf{K}_s \times \frac{d(\mathbf{K}_s^{-1})}{dt} \boldsymbol{\lambda}_{qd0}$$

speed voltage

Transforming Circuit Variables to ARF

$$\sin \cos \theta = \frac{\sin 2\theta}{2}$$

$$d\theta/dt = \omega$$

❖ Inductive Circuits:

$$\mathbf{K}_s \frac{d}{dt} [\mathbf{K}_s^{-1}] =$$

$$\mathbf{K}_s^{-1} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 1 \\ \cos(\theta - 120^\circ) & \sin(\theta - 120^\circ) & 1 \\ \cos(\theta + 120^\circ) & \sin(\theta + 120^\circ) & 1 \end{bmatrix}$$

$$\mathbf{K}_s = \left(\frac{2}{3} \right) \begin{bmatrix} \cos(\theta) & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin(\theta) & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

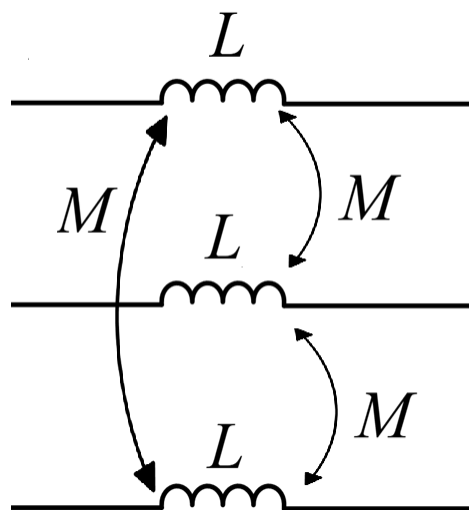
$$\frac{d}{dt} [\mathbf{K}_s^{-1}] = \omega \begin{bmatrix} -\sin \theta & \cos \theta & 0 \\ -\sin(\theta - 120^\circ) & \cos(\theta - 120^\circ) & 0 \\ -\sin(\theta + 120^\circ) & \cos(\theta + 120^\circ) & 0 \end{bmatrix}$$

$$\omega \left(\frac{2}{3} \right) \begin{bmatrix} 0 & 3/2 & 0 \\ -3/2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \omega \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\times \begin{bmatrix} \uparrow \uparrow \uparrow \\ \uparrow \uparrow \uparrow \\ \uparrow \uparrow \uparrow \end{bmatrix}$

Transforming Circuit Variables to ARF

❖ Inductive Circuits:



$$\mathbf{v}_{abc} = \frac{d\lambda_{abc}}{dt}$$

$$\lambda_{abc} = \mathbf{L}_{abc} \mathbf{i}_{abc}$$

$$\begin{cases} \mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc} \\ \mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0} \end{cases}$$

$$\mathbf{L}_{abc} = \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix}$$

$$\mathbf{K}_s \times \frac{d}{dt} (\mathbf{K}_s^{-1}) = \mathbf{W} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

❖ Flux Linkages:

Handwritten notes:

$$\hat{\mathbf{K}}_s^{-1} \lambda_{qd0} = \mathbf{L}_{abc} \hat{\mathbf{K}}_s^{-1} \mathbf{i}_{qd0} \rightarrow \mathbf{L}_{qd0}$$

$$\lambda_{qd0} = \mathbf{K}_s \mathbf{L}_{abc} \hat{\mathbf{K}}_s^{-1} \mathbf{i}_{qd0}$$

$$\lambda_{qd0} = \mathbf{L}_{qd0} \mathbf{i}_{qd0} \rightarrow \mathbf{L}_{qd0} = \mathbf{K}_s \mathbf{L}_{abc} \mathbf{K}_s^{-1} =$$

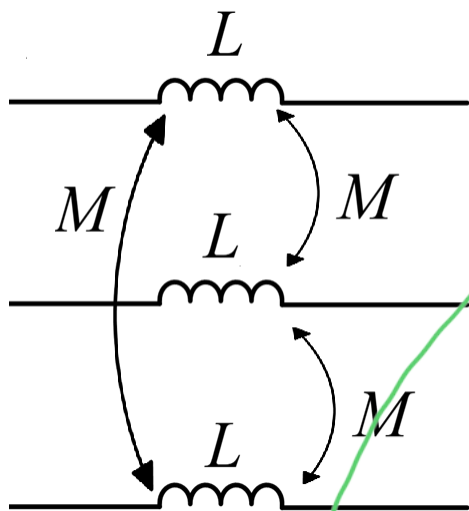
Handwritten notes:

$$\omega \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \lambda_{q0} \\ \lambda_{d0} \\ \lambda_{00} \end{bmatrix} = \omega \begin{bmatrix} \lambda_{d0} \\ -\lambda_{q0} \\ 0 \end{bmatrix}$$

$$\mathbf{L}_{qd0} = \begin{bmatrix} L_q & & \\ L - M & L_d & \\ & L - M & L_0 \\ & & L + 2M \end{bmatrix}_{12}$$

Transforming Circuit Variables to ARF

❖ Inductive Circuits:



$$\begin{cases} \mathbf{v}_{abc} = \frac{d\lambda_{abc}}{dt} \\ \lambda_{abc} = \mathbf{L}_{abc} \mathbf{i}_{abc} \end{cases}$$

$$\mathbf{L}_{abc} = \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix}$$

$$\begin{cases} \mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc} \\ \mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0} \end{cases}$$

$$\mathbf{L}_{qd0} = \begin{bmatrix} L_q & 0 & 0 \\ 0 & L_d & 0 \\ 0 & 0 & L_0 \end{bmatrix}$$

Define

$$\lambda_{dq0} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \lambda_{qd0} = \begin{bmatrix} \lambda_d \\ -\lambda_q \\ 0 \end{bmatrix}$$

$$\lambda_{qd0} = \mathbf{L}_{qd0} \mathbf{i}_{qd0}$$

$$\begin{aligned} \lambda_q &= L_q i_q \\ \lambda_d &= L_d i_d \\ \lambda_0 &= L_0 i_0 \end{aligned}$$

$$L_q = L_d = L - M$$

$$L_0 = L + 2M$$

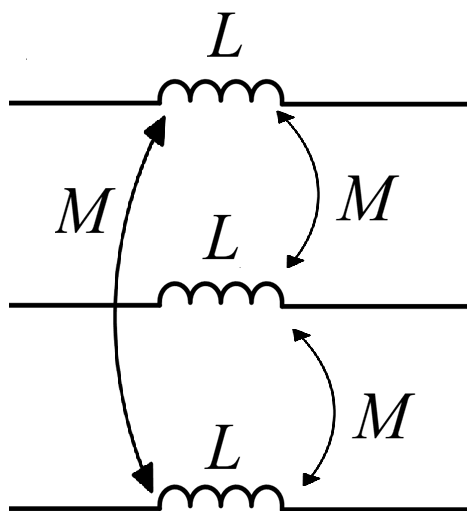
$$\mathbf{v}_{qd0} = \frac{d\lambda_{qd0}}{dt} + \omega \lambda_{dq0}$$

→ qd0

* speed voltage

Transforming Circuit Variables to ARF

❖ Inductive Circuits:



$$\begin{cases} \mathbf{v}_{abc} = \frac{d\boldsymbol{\lambda}_{abc}}{dt} \\ \boldsymbol{\lambda}_{abc} = \mathbf{L}_{abc} \mathbf{i}_{abc} \end{cases} \quad \mathbf{L}_{abc} = \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix}$$

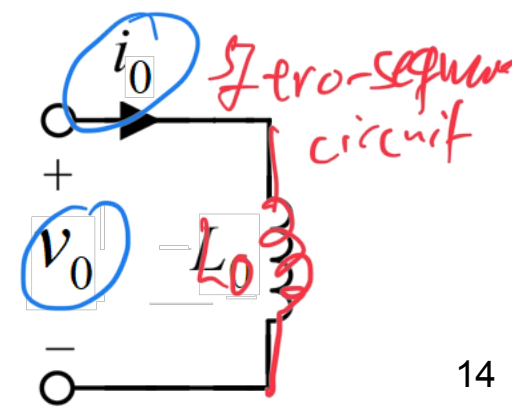
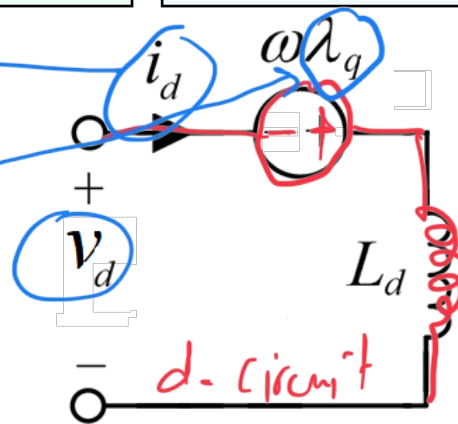
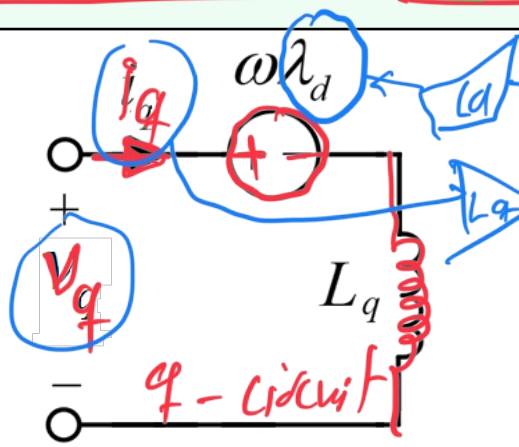
❖ Transformed Equivalent Circuit:

$$v_q = L_q \frac{di_q}{dt} + \omega \lambda_d \quad v_d = L_d \frac{di_d}{dt} - \omega \lambda_q \quad v_0 = L_0 \frac{di_0}{dt}$$

$$v_q = \frac{d\lambda_q}{dt} + \omega \lambda_d \quad v_d = \frac{d\lambda_d}{dt} - \omega \lambda_q \quad v_0 = \frac{d\lambda_0}{dt}$$

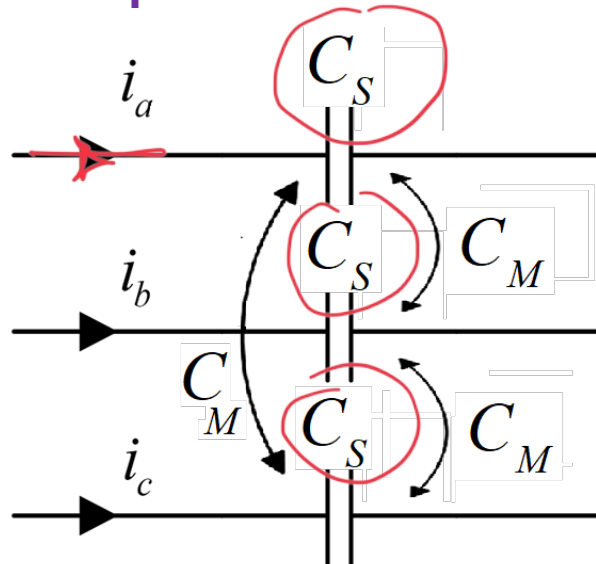
$$L_q = L_d = L - M \quad L_0 = L + 2M$$

$$\lambda_q = L_q i_q \quad \lambda_d = L_d i_d \quad \lambda_0 = L_0 i_0$$



Transforming Circuit Variables to ARF

❖ Capacitive Circuits:



$$\begin{cases} \mathbf{i}_{abc} = \frac{d\mathbf{Q}_{abc}}{dt} \\ \mathbf{Q}_{abc} = \mathbf{C}_{abc} \mathbf{V}_{abc} \end{cases}$$

$$\begin{cases} \mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc} \\ \mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0} \end{cases}$$

$$\mathbf{C}_{abc} = \begin{bmatrix} C_s & C_M & C_M \\ C_M & C_s & C_M \\ C_M & C_M & C_s \end{bmatrix}$$

Handwritten notes:

$$\mathbf{Q}_{qd0} = \begin{bmatrix} Q_q \\ Q_d \\ Q_0 \end{bmatrix}$$

$$\mathbf{Q}_{dq0} = \begin{bmatrix} Q_d \\ -Q_q \\ Q_0 \end{bmatrix}$$

Transform the variables:

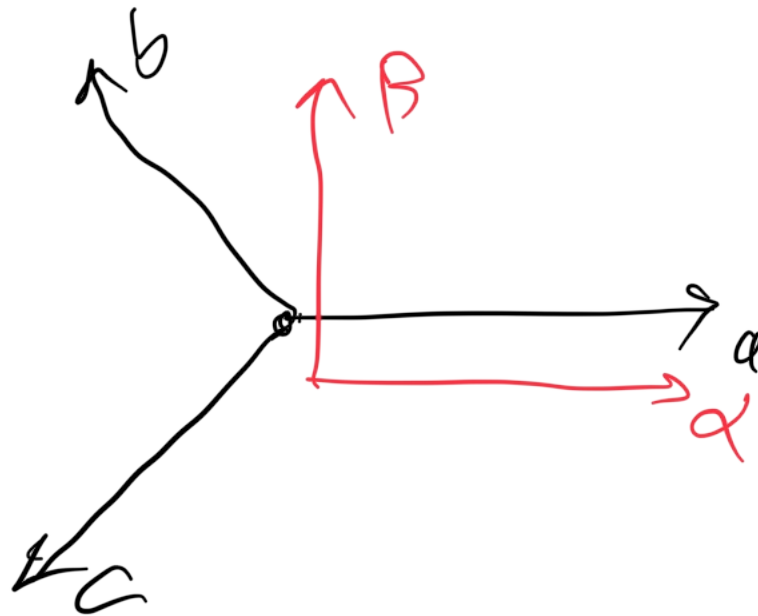
$$\mathbf{K}_s^{-1} \mathbf{i}_{abc} = \frac{d}{dt} [\mathbf{K}_s^{-1} \mathbf{Q}_{qd0}] \Rightarrow \mathbf{i}_{qd0} = \mathbf{K}_s \frac{d}{dt} [\mathbf{K}_s^{-1}] \mathbf{Q}_{qd0} + \frac{d\mathbf{Q}_{qd0}}{dt}$$

Define

$$\mathbf{Q}_{dq0} = \begin{bmatrix} Q_d & -Q_q & Q_0 \end{bmatrix}^T$$

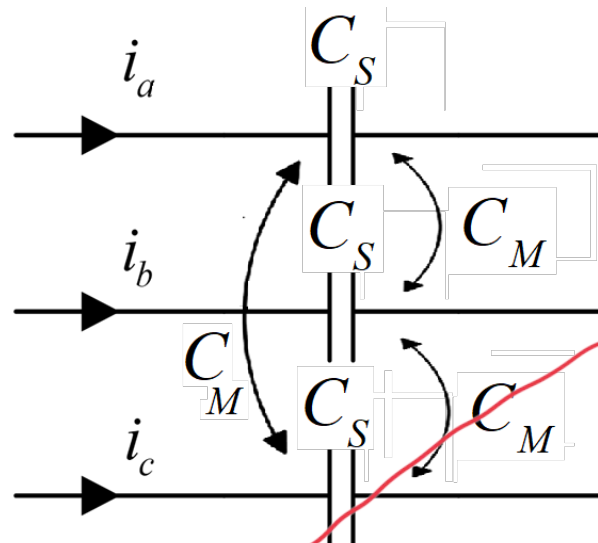
$$\mathbf{i}_{qd0} = \omega \mathbf{Q}_{dq0} + \frac{d\mathbf{Q}_{qd0}}{dt}$$

$$\dot{q}_{d0} = \omega \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{Q_d q_0} q_{d0} + \frac{d q_{d0}}{dt}$$



Transforming Circuit Variables to ARF

❖ Capacitive Circuits:



$$\mathbf{i}_{abc} = \frac{d\mathbf{Q}_{abc}}{dt}$$

$$\mathbf{Q}_{abc} = \mathbf{C}_{abc} \mathbf{V}_{abc}$$

$$\mathbf{C}_{abc} = \begin{bmatrix} C_S & C_M & C_M \\ C_M & C_S & C_M \\ C_M & C_M & C_S \end{bmatrix}$$

$$\begin{cases} \mathbf{f}_{qd0} = [\mathbf{K}_s] \mathbf{f}_{abc} \\ \mathbf{f}_{abc} = [\mathbf{K}_s^{-1}] \mathbf{f}_{qd0} \end{cases}$$

Handwritten notes:

$$\begin{aligned} Q_q &= C_q V_q \checkmark \\ Q_d &= C_d V_d \checkmark \\ Q_o &= C_o V_o \\ C_{qd0} \end{aligned}$$

❖ Electric Charges:

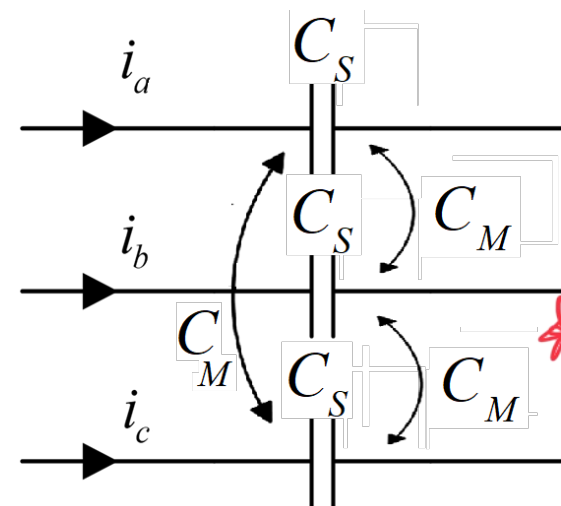
$$K_s^{-1} Q_{qd0} = C_{abc} \cdot K_s^{-1} V_{qd0}$$

$$Q_{qd0} = K_s C_{abc} \cdot K_s^{-1} V_{qd0}$$

$$\mathbf{Q}_{qd0} = \mathbf{C}_{qd0} \mathbf{V}_{qd0} \Rightarrow \mathbf{C}_{qd0} = \mathbf{K}_s \mathbf{C}_{abc} \mathbf{K}_s^{-1} = \begin{bmatrix} C_S - C_M & C_S - C_M & C_S + 2C_M \\ C_S - C_M & C_S - C_M & C_S + 2C_M \\ C_S + 2C_M & C_S + 2C_M & C_S - C_M \end{bmatrix}$$

Transforming Circuit Variables to ARF

❖ Capacitive Circuits:



$$\begin{cases} \mathbf{i}_{abc} = \frac{d\mathbf{Q}_{abc}}{dt} \\ \mathbf{Q}_{abc} = \mathbf{C}_{abc} \mathbf{V}_{abc} \end{cases} \quad \mathbf{C}_{abc} = \begin{bmatrix} C_S & C_M & C_M \\ C_M & C_S & C_M \\ C_M & C_M & C_S \end{bmatrix}$$

❖ Transformed Equivalent Circuit:

** $i_q = C_q \frac{dv_q}{dt} + \omega Q_d$*

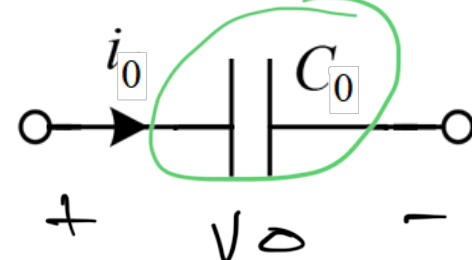
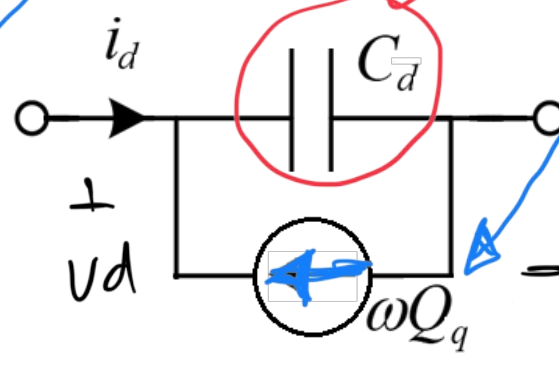
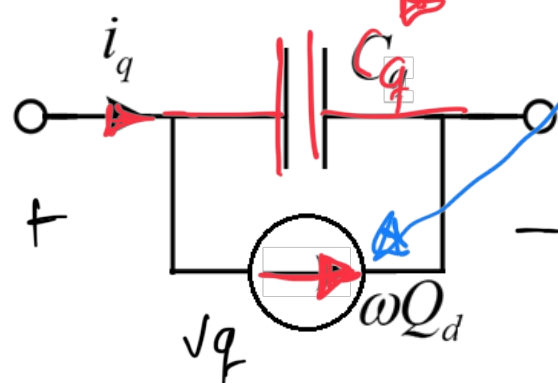
$i_d = C_d \frac{dv_d}{dt} - \omega Q_q$

$i_0 = C_0 \frac{dv_0}{dt}$

$$i_q = \frac{dQ_q}{dt} + \omega Q_d \quad i_d = \frac{dQ_d}{dt} - \omega Q_q \quad i_0 = \frac{dQ_0}{dt}$$

$$\begin{aligned} C_q &= C_d = C_S - C_M \\ C_0 &= C_S + 2C_M \end{aligned}$$

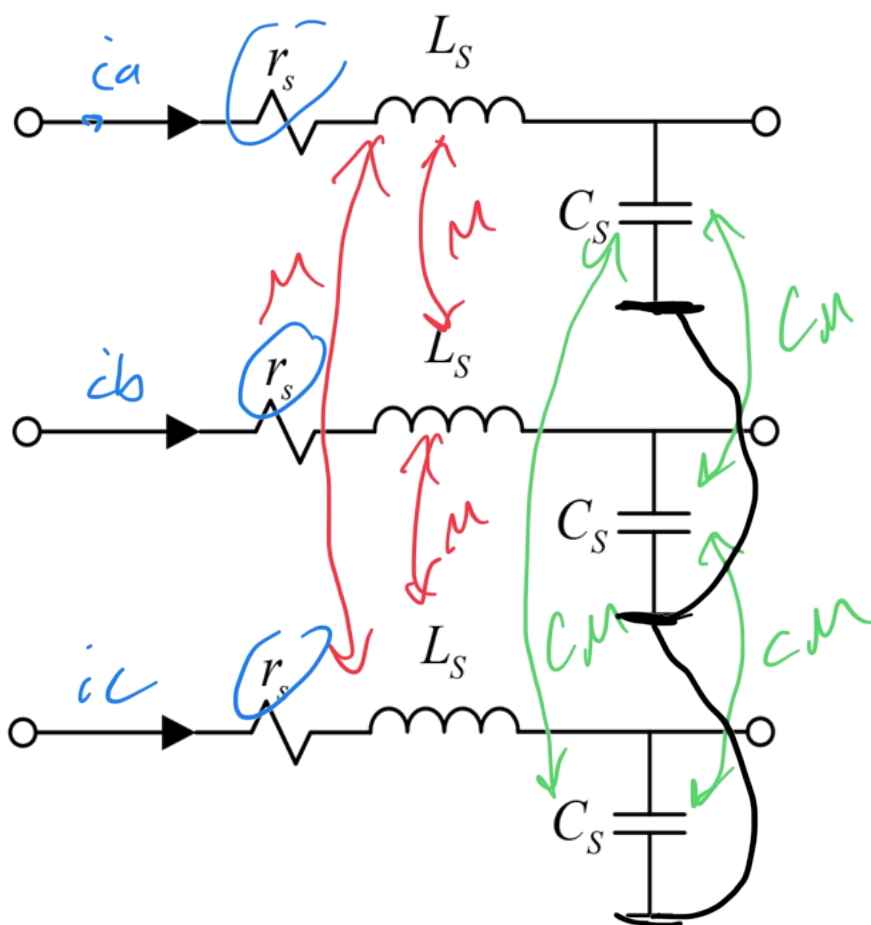
$$Q_q = C_q v_q \quad Q_d = C_d v_d \quad Q_0 = C_0 v_0$$



Transforming Circuit Variables to ARF

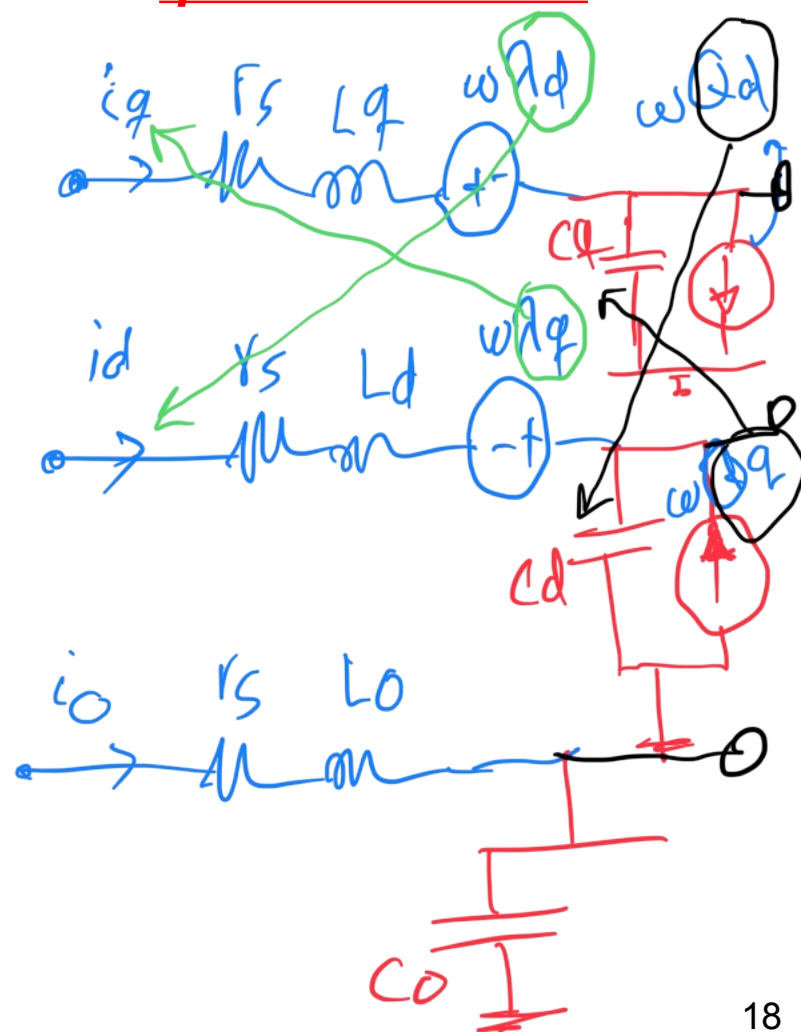
❖ **RLC Circuits:**

❖ **abc coordinates:**



$$L_d = L_q = L_s - M \quad L_o = L + 2M$$

❖ **qd0 coordinates:**



Selection of Reference Frame

❖ Special Cases of ARF:

فابو

Name of Reference Frame	Speed of Frame	Transformation Matrix	Variables
Arbitrary	ω	$\mathbf{K}_s$	$\mathbf{f}_{qd0}$
Stationary α/β	0	$\mathbf{K}_s^s$	$\mathbf{f}_{qd0}^s$
Synchronous	ω_e	$\mathbf{K}_s^e$	$\mathbf{f}_{qd0}^e$
Rotor	ω_r	$\mathbf{K}_s^r$	$\mathbf{f}_{qd0}^r$



- ✓ The speed as well as the angle of the reference frame can be selected differently to achieve different objectives

Reference Frames for 3-Phase Systems

❖ Transformation Between Reference Frames:

- Assume two different *qd* reference frames x with angle θ_x and y with angle θ_y

$$\mathbf{f}_{qd}^y = {}^y\mathbf{K}^x \mathbf{f}_{qd}^x$$

$$\begin{aligned}\theta_n &= \theta_n(0) + \int \omega_n dt \\ \theta_y &= \theta_y(0) + \int \omega_y dt\end{aligned}$$

$${}^y\mathbf{K}^x = \begin{bmatrix} \cos(\theta_y - \theta_x) & -\sin(\theta_y - \theta_x) \\ \sin(\theta_y - \theta_x) & \cos(\theta_y - \theta_x) \end{bmatrix}$$

$$\mathbf{f}_{qd}^x = {}^x\mathbf{K}^y \mathbf{f}_{qd}^y$$

$${}^x\mathbf{K}^y = [{}^y\mathbf{K}^x]^{-1} = [{}^y\mathbf{K}^x]^T$$

